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SUPERSEDING
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April 23, 1958

FEDERAL SPECIFICATION

CLOTH, COTTON, BUCKRAM

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for cloth, cotton, buckram, primarily intended for use in the manufacturing of binders for books and stiffening material for clothing (see 6.1).

1.2 Classification.

1.2.1 *Types and classes.* Cloth, cotton, buckram shall be of the following types and classes, as specified (see 6.2).

Type I.—For bookbinding.
- Class 1.—4.7 ounces per square yard—light.
- Class 2.—5.7 ounces per square yard—medium.
- Class 3.—7.9 ounces per square yard—heavy.

Type II.—For clothing.
- Class 1.—3.8 ounces per square yard.
- Class 2.—4.5 ounces per square yard.
- Class 3.—12.0 ounces per square yard.
- Class 4.—17.0 ounces per square yard.

Type III.—For clothing, double texture (2 plies of type II, class 3).

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

Federal Specifications:

CCC-T-191—Textile Test Methods.

PPP-P-51—Packaging, Packing and Marking of Textile Fabrics (Woolens, Worsteds, Cottons, Silks and Synthetics).

Federal Standards:

Fed. Std. No. 4—Glossary of Fabric Imperfections.

Fed. Std. No. 102—Preservation, Packaging and Packing Levels.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Standard:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Rules and Regulations Under the Textile Fiber Products Identification Act, Effective March 3, 1960.

(Copies may be obtained free of charge from the Federal Trade Commission, Washington 25, D. C.)

3. REQUIREMENTS

3.1 Standard sample. When a standard sample is available, the dyed cloth shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 Material. The material for manufacture of the yarn shall be cotton and shall be cleaned, carded, drawn, and evenly spun.

3.3 Color. The cloth shall be natural, dyed, or bleached as specified (see 6.2). The color shall be produced by the use of such colorants and such methods that may be specified by the contracting officer. The use of sulfur dyes or substances containing elementary sulfur compounds capable of oxidation to sulfuric acid is prohibited. Tests for sulfur shall be conducted as specified in 4.4.

3.3.1 *Matching.* The color shall match the standard sample (see 6.3) under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin, and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin.

3.3.2 *Colorfastness.* The dyed cloth shall show fastness to water, crocking, and light equal to or better than the standard sample (see 6.3). When the standard sample is not referenced for colorfastness, the dyed cloth shall show "good" fastness to water, crocking, and light when tested as specified in 4.4.

TABLE I.—*Physical requirements for type I*

Class	Width exclusive of selvage, minimum[1]	Weight per square yard		Number of yarns per inch, minimum		Breaking strength, minimum	
		Stripped[2] (minimum)	Finished (maximum)	Warp	Filling	Warp	Filling
	Inches	*Ounces*	*Ounces*			*Pounds*	*Pounds*
1	38	4.7	6.5	30	25	65	35
2	38	5.7	8.5	82	21	100	55
3	38	7.9	10.0	82	24	120	70

[1] Selvages shall not exceed ¼ inch.

[2] The basis of weight comparison shall be the weight of the cloth which remains after the finished cloth has been stripped of all starch and loading, or impregnating materials.

TABLE II.—*Physical requirements for type II*

Class	Width exclusive of selvage, minimum[1]	Finished weight per square yard, minimum	Number of yarns per inch, minimum		Breaking strength, minimum		Sizing and other non-fibrous materials, maximum
			Warp	Filling	Warp	Filling	
	Inches	*Ounces*			*Pounds*	*Pounds*	*Percent*
1	35½	3.8	46	42	50	40	35
2	32	4.5	50	42	60	50	60
3	34	12.0	64	20	150	80	50
4	34	17.0	74	21	130	40	65

[1] Selvages shall not exceed ¼ inch.

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3.4 Weave. The weave shall be plain, except that type I, classes 2 and 3 shall be plain with 2 ends woven as one; and type II, class 3 shall be a 4-harness, 2 up and 2 down herringbone twill, 20 warp ends right and 20 warp ends left.

3.5 Type I, for bookbinding.

3.5.1 *Physical requirements.* Type I cloth shall conform to the requirements in table I when tested as specified in 4.4.

3.5.2 *Finish.* The cloth shall have a starch finish.

3.5.3 *Length.* Unless otherwise specified (see 6.2), type I cloth shall be furnished in rolls of one continuous piece of not less than 40 yards.

3.6 Type II, for clothing.

3.6.1 *Physical requirements.* Type II cloth shall conform to the requirements in table II when tested as specified in 4.4.

3.6.2 *Finish.*

3.6.2.1 *Class 1.* Class 1 cloth shall have a starch swing finish produced by a shifting tenter frame which breaks the adhesion of the finish at the interstices. Determination of the swing finish shall be made as specified in 4.4.

3.6.2.2 *Class 2.* Class 2 cloth shall have a smooth permanent finish suitable for lining shirt collars. The material shall show no discoloration, chlorination, degradation, or loss of more than 10 percent of the original breaking strength of the cloth when tested as specified in 4.4.

3.6.2.3 *Classes 3 and 4.* Classes 3 and 4 cloth shall have a starch finish.

3.6.3 *Shrinkage.* Type II, class 2 cloth shall not shrink more than one percent in either the warp or filling direction when tested as specified in 4.4 The preshrinking process used shall not be identified by name or trademark, either on the cloth, ticket, or the package.

3.6.4 *Length.* Unless otherwise specified (see 6.2), type II cloth shall be furnished in rolls of one continuous piece of not less than 50 yards.

3.7 Type III, double texture for clothing.

3.7.1 *Construction.* Type III cloth shall be made of two plies of type II, class 3 cloth and shall be joined by laminating the two plies together.

3.7.2 *Laminating material.* The laminating material shall be of a synthetic resin type, unaffected by standard dry cleaning solvents and standard launderings. There shall be no objectionable odor caused by these materials.

3.7.2.1 *Thickness.* The thickness of the laminating material shall be not less than 0.014 inch nor more than 0.016 inch when tested as specified in 4.4.

3.7.3 *Breaking strength.* The breaking strength of the laminated cloth shall be not less than that required for a single ply of type II, class 3 cloth when tested as specified in 4.4.

3.7.4 *Stiffness.* Type III cloth shall have an average stiffness of not less than 0.400 inch-pounds for both warp and filling when tested as specified in 4.4.

3.7.5 *Resistance to dry cleaning.* The laminated cloth shall show no evidence of delamination when subjected to three cycles of dry cleaning when tested as specified in 4.4.

3.7.6 *Resistance to laundering.* The laminated cloth shall show no evidence of delamination when subjected to six cycles of laundering when tested as specified in 4.4.

3.7.7 *Length.* Unless otherwise specified (see 6.2), type III cloth shall be furnished in rolls of one continuous piece of not less than 40 yards.

3.8 Fiber identification. Each roll shall be labeled, ticketed or invoiced for fiber content in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act effective March 3, 1960.

3.9 Workmanship. The finished cloth shall conform to the quality and grade of product established by this specification. Occurrence of

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defects shall not exceed the applicable acceptable quality levels.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Contractor's inspection responsibility. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other inspection facilities and servces acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Examination of the end item. Inspection shall be in accordance with Military Standard MIL–STD–105 except where otherwise indicated. The cloth shall be examined to determine conformance to this specification. Defects found shall be classified in accordance with 4.2.1, 4.2.2, and 4.2.3.

4.2.1 *Yard-by-yard examination.* The required yardage of each piece shall be inspected on the face and visual defects classified as listed in table III. The defects found shall be counted regardless of their proximity to each other, except where two or more defects represent a single condition of the cloth in which case only the more serious defect shall be counted. A continuous defect shall be counted as one defect for each warpwise yard or fraction thereof in which it occurs. The unit of product for this examination shall be 1 linear yard. The inspection level shall be level I of Military Standard MIL–STD–105. The acceptable quality level shall be 4.0 defects per hundred units (yards). The lot size shall be expressed in units of 1 yard each. The number of pieces from which the sample is to be selected shall be in accordance with table IV. An approximately equal number of yards shall be examined in each piece in the sample. Not more than one roll shall be taken from any container.

TABLE III.

Fed. Std. No. 4	Defects
90, 91, 92	Hole, cut, or tear (including selvage)—any.
11, 33	Broken or missing yarns—two or more contiguous, 2 inches or more missing.
9, 12, 35	Open place, crack or hitchback, opening at widest point four or more normal yarn diameters or sequence of openings resulting in weak place.
13, 34	Floats or skips, multiple—½ inch or more in combined warp and filling directions.
9	Light place or fine filling bar.[1]
3, 5, 27, 77	Slough off, jerked-in filling, slugs, slubs or gouts—forming a raised surface above the normal plane of the cloth.[1]
6, 7, 28, 29	Kink, knots or untrimmed ends—accumulation.[1]
78	Smash—any.
89	Tender or weak place.
74, 75, 76[2], 69, 70	Spot, stain, or streak—uncleaned ½ inch or more in combined directions.
59, 61	Crease or wrinkle, hard embedded.

[1] Clearly visible at normal inspection distance (approximately 3 feet).

[2] Except defects of spots, stains, or streaks shall be counted regardless of cause i. e. not restricted to rust, oil, or grease.

4.2.2 *Overall examination.* The pieces for this examination shall be those selected for yard-by-yard examination. Each defect listed below shall be counted not more than once in each piece examined. The unit of product shall be one piece. The acceptance number (maximum number of defects acceptable) shall be as shown in table IV for the appropriate number of pieces in the sample.

Defects

Shade (when colors are specified):
Off shade, shaded end to end, side to side,

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side to center or throughout the piece.
Cloudy, mottled or streaky throughout the piece.
Uncleanness.
Objectionable odor.
Width less than specified.
Preshrinking process dentified by name or trademark.

TABLE IV.—*Sampling table*

Lot size in yards	Sample size in pieces	Acceptable number
Up to 1,300[1]	3	0
1,301 to 3,200	5	0
3,201 to 8,000	7	0
8,001 to 22,000	10	0
22,001 to 110,000	15	0
110,001 and over	25	1

[1] If lot contains less than three pieces, each piece in the lot shall be examined.

4.2.3 *Examination for length.* The pieces for this examination shall be those selected for yard-by-yard examination.

4.2.3.1 *Individual pieces.* Each piece in the sample shall be examined for gross length. Any gross length found to be less than the specified minimum length, or any gross length found to be two or more yards below the gross length marked on the piece ticket, shall be considered as a defect with respect to length. The unit of product shall be one piece. The acceptance number (maximum number of defects acceptable) shall be as shown in table IV for the appropriate number of pieces in the sample.

4.2.3.2 *Total yardage in sample.* The lot shall be unacceptable if the total of the actual gross lengths of pieces in the sample is less than the total of the gross lengths marked on the tickets.

4.3 Examination of preparation for delivery requirements. An examination shall be made to determine that packaging, packing and marking required by section 5 of this specification are complied with. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be sealed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be L–4 and the AQL shall be 4.0 defects per hundred units.

Examination	Defect
Markings (Exterior and interior)	Incorrect; omitted; illegible; of improper size, location, sequence, or method of application. Preshrinking process identified by name or trademark.
Materials	Any component missing. Any component damaged, affecting serviceability.
Workmanship	Inadequate application of components such as: incomplete closure of case liners, container flaps, loose strapping, etc.
	Bulging or distortion of containers.
Weight	Gross or net weight exceeds requirement.

4.4 Testing of the end item. The methods of testing specified in Federal Specification CCC–T–191, wherever applicable, and as listed in table VI shall be followed. The physical and chemical values specified in section 3 apply to the average of the determinations made on a unit of product for test purposes as specified in the applicable test method. The unit of product shall be 1 continuous yard, full width, of the finished cloth except that the unit of product for type II, class 2 cloth shall be 3 yards, full width, of the finished cloth. The sample size shall be in accordance with table V. The lot shall be

TABLE V.

Lot size (yards)	Sample size
800 or less	2
801 to 22,000	3
22,001 and over	5

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TABLE VI.—*Test methods*

Characteristics	Requirement reference	Test method	Number of determinations per unit of product	Results reported as:
Weight, oz. sq. yd.	Tables I and II	5041	..	.....
Yarns per inch	Tables I and II	5050	..	.....
Breaking strength:				
Warp	Tables I and II	(see 4.4.1)	6	Average of 6 determinations to the nearest 1.0 lb.
Filling	Tables I and II	(see 4.4.1)	6	Average of 6 determinations to the nearest 1.0 lb.
Weave	3.4	Visual	1	Pass or fail
Swing finish	3.6.2.1	(see 4.4.3)	1	Pass or fail
Chlorine degradation:				
Warp	3.6.2.2	(see 4.4.2)	8	Compute to nearest 0.1 percent
Filling	3.6.2.2	(see 4.4.2)	8	Compute to nearest 0.1 percent
Thickness	3.7.2.1	5030	..	.....
Stiffness	3.7.4	5202	..	
Colorfastness to:				
Light[1]	3.3.2	5660	..	.....
Water	3.3.2	5630	..	.....
Crocking	3.3.2	5651	..	.....
Sulfur	3.3	(see 4.4.4)	1	Pass or fail
Nonfibrous material:				
Type II, classes 1, 3 and 4	Table II[2]	2611	..	.....
Type II, class 2 (permanent finish)	Table II[3]	Accept on certificate of compliance	..	.....
Shrinkage in laundering[4]	3.6.3	5550	..	.....
Dry cleaning (3) cycles)	3.7.5	5622	1	Pass or fail
Laundering (6 cycles)	3.7.6	5550	1	Pass or fail

[1] In the event of a dispute resulting from the test with method 5660, or as a result of suspected anomalous behavior of certain dye types or formulations, the contracting officer may authorize the exposure to natural light in accordance with method 5662.

[2] In lieu of test results, the contractor may furnish a certificate of compliance with each lot or shipment stating that the nonfibrous material in the cloth does not exceed specified limits.

[3] The contractor shall submit a certificate of compliance with each lot stating that the material has a permanent type finish suitable for lining shirt collars and that the nonfibrous materials including resins, plasticizers and other substances do not exceed 60 percent by weight of the bone dry weight of the finished cloth.

[4] The three specimens for this test shall be selected from the cloth (unit of product) as follows: one specimen from each side of the cloth within 3 inches of the selvage and one specimen from the center of the cloth. No two specimens shall contain the same filling yarns. Each elongation value shall be prefixed with a minus sign and both enclosed in parenthesis.

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unacceptable if one or more units fail to meet a test requirement. When the data in the "Number of determinations per unit of product" and the "Results reported as" columns are not specified in table VI, they shall be as required by the referenced method.

4.4.1 *Breaking strength.*

4.4.1.1 Type I, classes 1, 2, and 3, and type II, class 4, shall be tested for breaking strength in accordance with method 5102 of Federal Specification CCC–T–191, except that the specimens shall be obtained as follows: A strip of cloth 7 inches long in the warp direction shall be cut across the entire width of the cloth. Twelve specimens shall be cut from this strip, six with the long dimension in the direction of the warp and six with the long dimension in the direction of the filling. Each specimen shall be 1 inch in width. No two specimens shall contain the same yarns and care should be taken to insure that the yarns to be subjected to test have not been damaged during the cutting of the specimens.

4.4.1.2 Type II, classes 1, 2, and 3 shall be tested for breaking strength in accordance with method 5100 of Federal Specification CCC–T–191.

4.4.2 *Test for deleterious effect of finish on breaking strength (type II, class 2).* A piece of cloth, not less than 20 inches long and full width, shall be laundered in accordance with method 5550 of Federal Specification CCC–T–191 and shall then be treated for one hour with 0.7 percent available chlorine solution. The piece shall be rinsed in a sour solution and washed thoroughly until a control piece shows no trace of chlorine. The piece shall be air dried and then conditioned for 4 hours in an atmosphere having a relative humidity of 65 ± 2 percent and a temperature of 75° ± 5°F. Eight pairs of specimens, four warpwise and four fillingwise, with each pair having the same yarns shall be taken for tests. Four warp specimens and four filling specimens shall be tested in a pendulum-type machine in accordance with method 5100 of Federal Specification CCC–T–191. The remaining warp and filling specimens shall be oven dried at 105° to 110°C. for two hours and tested in the same manner as the first eight specimens. The difference in the average breaking strength of the respective warp and filling samples shall be computed and the percentage of loss determined. If a specimen slips or breaks in or at the edge of the jaws, the result shall be discarded and another pair of specimens taken, both having identical warp and filling yarns and the test repeated on this pair of specimens.

4.4.3 *Swing finish.* A piece of type II, class 1 cloth 10 by 10 inches shall be folded on a 45-degree bias without creasing. Holding the ends of the fold with both hands and exerting a pull the cloth will stretch or buckle. Cloth having a swing finish will stretch or buckle.

4.4.4 *Test for presence of sulfur.* Dissolve 100 grams of stannous chloride crystals in 100 ml. of hydrochloric acid (35 percent concentration) and add 50 ml. of distilled water. To preclude interference in dye identification, the sample first shall be boiled in distilled water to remove any trace of chemical used in the dyeing process. One to 1.5 grams of the finished material, cut into small pieces, shall be used for the test. Place the sample in a test tube of suitable size and cover with the stannous chloride solution. Place a filter paper which has been moistened with one drop of lead-acetate solution (made by dissolving 5 grams of reagent-grade lead acetate in 100 cc. of distilled water; add a drop of glacial acetic acid to clear up if cloudy), over the mouth of the test tube. (Punch needle hole in paper.) Heat slowly over a low flame until the solution is boiling. Solution shall not be boiled for more than 10 to 15 seconds as prolonged boiling will reduce the sensitivity of the test. A brown to black stain appearing on the paper indicates the presence of sulfur.

5. PREPARATION FOR DELIVERY

(Federal Standard No. 102 shall be referred to for definitions and application of the various levels of packaging and packing for civilian procurements.)

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5.1 Packaging. Packaging shall be level A, B, or C as specified (see 6.2).

5.1.1 *Levels A, B, and C.* The rolls of cloth shall be packaged in accordance with the applicable requirements for class 3 fabrics of Federal Specification PPP–P–51.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 *Levels A, B, and C.* The rolls of cloth, packaged as specified in 5.1.1, shall be packed in accordance with the applicable requirements of Federal Specification PPP–P–51.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the requirements of Federal Specification PPP–P–51.

6. NOTES

6.1 Intended use. Type I cloth (for bookbinding) is used for textbooks, edition, trade, catalog, and similar uses, except that it should not be used for library bindings. Types II and III cloth are used as interlining for clothing. Type II, class 2 cloth is suitable for lining shirt collars and type III cloth (2 ply) is used when a high degree of stiffness is desired as in Shoulder Marks for cadet uniforms.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Type and class of cloth required (see 1.2).
(*c*) Color, if dyeing is required, or whether natural or bleached (see 3.3).
(*d*) Length, if other than specified (3.5.3, 3.6.4, and 3.7.7).
(*e*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2.1 In the preparation of contracts or orders, it must be noted that various options, choices, and alternatives, as indicated in Federal Specification PPP–P–51 may be exercised in the preparation for delivery of this commodity.

6.3 Standard sample. For access to standard sample, address the procuring office issuing the invitation for bids.

6.4 Suggested basic cloths for type I buckram are as follows:

Class 1—42 inches, 36 by 32, 2.60 yards per lb. clear Osnaburg.
Class 2—41 inches, 84 by 26, 2 yards per lb. SF Flat Duck.
Class 3—42 inches, 84 by 28, 1.45 yards per lb. of DF Flat Duck.

6.5 Transportation description. The transportation descriptions and minimum weights applicable to this commodity are:

Rail:
Buckram, cotton.
Carload minimum weight—Any quantity.
Motor:
Buckram, cotton (Specify single ply or other than single ply).
Motor volume minimum weight—Any quantity.

6.6 Level B packaging and packing shall be used by ordering Agencies only when shipment is known to be made to a holding Supply Depot or Warehouse from which partial shipments will be made to ultimate consumers.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise, as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS
Army—Q
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